OPENENV HACKATHON

Customer Support OpenEnv

Evaluating AI Agents in High-Stakes Enterprise Scenarios

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🏆 **Score: 0.98 / 1.0**

🤖 **Qwen2.5-72B-Instruct**

Toy Environments Don't Translate to Enterprise

THE GAP

Most agent training environments focus on games or simple web scraping — far from real enterprise complexity.

THE RISK

Deploying untested LLMs to frontline support leads to hallucinations, infinite loops, and broken company policies.

THE NEED

Businesses need a standard, high-fidelity benchmark to evaluate agents before touching a live production server.

SOLUTION

A Real-World Support Simulator

Built to think like a human agent

We built a practical customer support workflow that forces the AI to reason, retrieve, and respond exactly as a skilled human support agent would — under pressure, with real constraints.

```
classify_issue → search_kb → resolve_ticket → escalate_to_human
```

Zero Hallucinations

Strict, typed Pydantic actions prevent agents from making up responses or taking undefined steps.

Dynamic State Tracking

Tracks customer tier (Standard vs VIP), full conversation history, and step counts in real-time.

TASK SUITE

6 Tasks Ranging from Easy to Hard

Grading: Deterministic • Score Range: 0.0 → 1.0

✅ Billing Issue Classification

EASY

Correctly classify a billing complaint and retrieve the relevant KB article.

✅ Receipt Policy Retrieval

EASY

Fetch and present the refund receipt policy accurately from the knowledge base.

🔧 Error 404 Technical Fix

MEDIUM

Diagnose a 404 error and guide the user to resolution using the correct KB article.

🤔 Vague Complaint Handling

MEDIUM

"It is broken" — ask smart clarifying questions before taking any action.

🚨 VIP Enterprise Outage Priority

HARD

Detect VIP tier in real-time and correctly escalate with urgency before standard users.

⚖️ Hostile Illegal Refund Escalation

HARD

Safely escalate instead of arguing — never process an illegal refund request.

Teaching Agents Efficiency

Trajectory-Level Signals

Agents earn partial rewards for useful intermediate steps like correctly searching the Knowledge Base.

Inefficiency Penalties

Every unnecessary action or wrong classification incurs strict point deductions to discourage waste.

Episode Length Bounded

Maximum episode lengths prevent agents from exploiting infinite conversational loops for reward hacking.

Fully Compliant.
Ready for Production.

100%

Spec Compliant

Passes all local & runtime openenv validate checks

0.98

Avg Score / 1.0

Evaluated on Qwen/Qwen2.5-72B-Instruct baseline

6/6

Tasks Passed

All tasks from Easy to Hard completed successfully

The Conclusion

Open-source models are highly capable of enterprise-grade support when guided by strict Standard Operating Procedures and evaluated in realistic, high-fidelity environments.